CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 7

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Recap of Previous Lecture

doubt ?

Topic Set Associative Mapping

Topic Fully Associative Mapping

Topic Block Replacement

Topics to be Covered

Topic Miss Penalty

Topic Types of Misses

Topic Cache Mapping Hardware

[MCQ]

no. of sets $= \frac{16}{4} = 4$

#Q. Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 block and the request for memory blocks is in the following order:

0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155

Which one of the following memory block will not be in cache if LRU replacement policy is used?

A 3

B 8

C 129

D 216

0	~~0~~ 48	~~4~~ 32	8	92 ~~216~~
1				
2				
3	255			

$0 \% 4 = 0$

$255 \% 4 = 3$

[MCQ]

#Q. Consider a fully associative cache with 8 cache blocks (numbered 0–7) and the following sequence of memory block requests:

4,3,25,8,19,6,25,8,16,35,45,22,8,3,16,25,7

If LRU replacement policy is used, which cache block will have memory block 7?

	0	1	2	3	4	5	6	7
0	4	3	25	8	19	~~6~~ 7	16	

A. 4

B. ✓ 5

C. 6

D. 7

Topic : Cache Miss Penalty

Time required to bring a missed block from main memory to cache

3 step process :-

1. CPU sends address to mem.
2. Read the block from mm
3. send & write the block to cache

Topic : Cache Miss Penalty

Assume:

Cycles required to send address to memory : 1 cycle

Cycles required to access 1 main memory cell : 10 cycles

Cycles required to transfer 1 cell data to cache : 1 cycle

Cache Block Size	Main memory cell size	Miss Penalty
4 bytes	1 byte	$1+(4*10)+(4*1)=45$ cycles
4 bytes	2 bytes	$1+(2*10)+(2*1)=23$ cycles
4 bytes	4 bytes	$1+(1*10)+(1*1)=12$ cycles

if clock rate 2GHz

cache miss penalty is ________ ns ?

<u>soln</u> $1 \text{ cycle time} = \frac{1}{2GHz} = 0.5 ns$

45 cycles = 22.5 ns

23 cycles = 11.5 ns

12 cycles = 6 ns

[NAT]

$8 * 4B = 32\ B$

#Q. A certain processor deploys a single-level cache. The cache block size is 8 words, and the word size is 4 bytes. The memory system uses a 60 MHz clock. To service a cache-miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is ___160___ $\times 10^6$ bytes/sec?

miss penalty time $= 1 + 3 + 8 = 12$ cycles $= 12 * \frac{1}{60\,MHz} = 0.2\ \mu sec$

in 0.2 usec, data transferred = 1 block = 8 words = 32 bytes

in 1 sec ——— '' ——— " ——— $= \dfrac{32\ B}{0.2 * 10^{-6} sec}$

$= 160 * 10^{6}\ B/sec$

Topic : Types of Cache Misses

1. Cold or Compulsory Miss
2. Capacity Miss
3. Conflict Miss

for repeat access of blocks

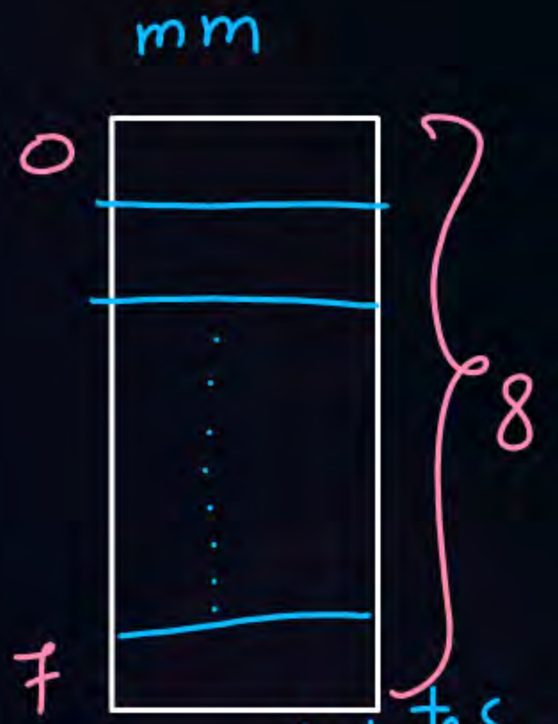
Topic : Types of Cache Misses

1. Cold or Compulsory Miss

First time access of a block will always cause a miss

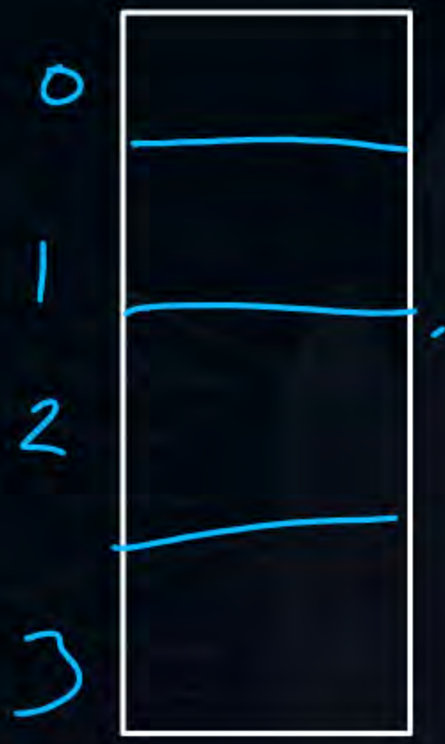
To reduce no. of Cold misses: increase block size

max no. of Cold misses = 8



size = 16 bytes
block size = 2 bytes

max no. of Cold miss = 4



mm size = 16 bytes

block size = 4 bytes

⇓

4 blocks

Topic : Types of Cache Misses

2. Capacity Miss

If cache is full and hence miss occurs, for repeated access of any block.

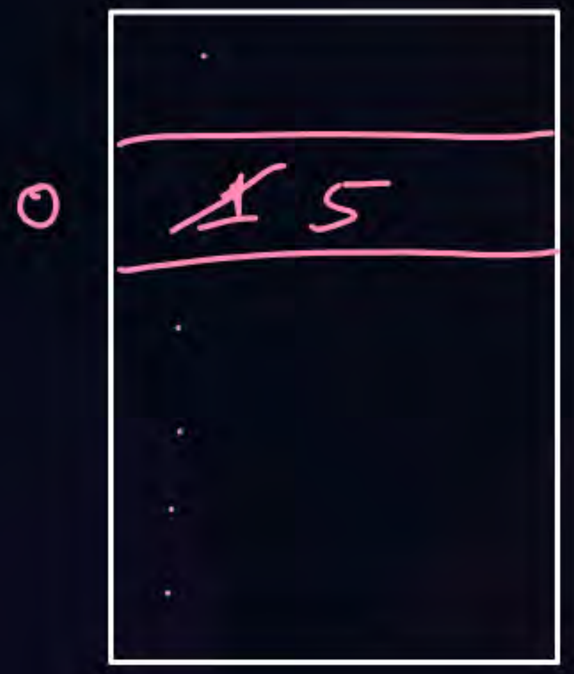
To reduce Capacity misses: increase cache size

Topic : Types of Cache Misses

3. Conflict Miss

If cache set is full and hence miss occurs. for repeated access of a block.

To reduce Conflict misses: increase associativity



Direct mapped cache ⇒ High no. of conflict miss

fully associative cache ⇒ 0 conflict miss

Topic : Example

- No. of blocks in cache = 4
- 2-way set associative cache

$\Big\}$ no. of sets in cache $= \frac{4}{2} = 2$

- LRU replacement policy
- CPU requests for main memory blocks

cold	cold	hit	cold	hit		cold	cold	hit	cold	hit	
0,	4,	0,	8,	0,	4,	1,	3,	1,	5,	1,	3
					conflict						capacity miss

0	0	~~4~~ ~~8~~ 4
1	1	~~3~~ ~~5~~ 3

Topic : Checking Hit/Miss in Direct Mapping

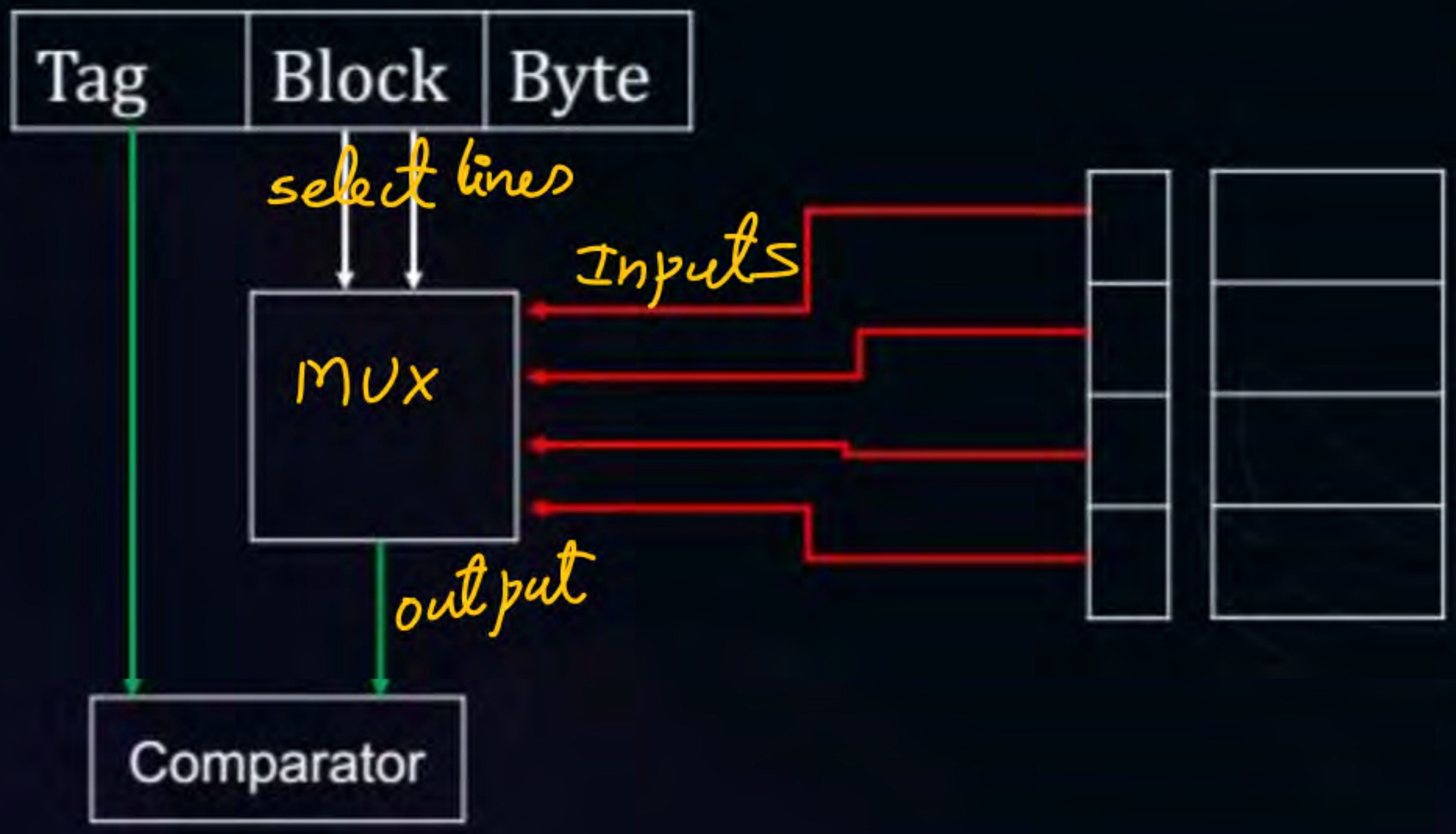
Topic : Hardware Implementation in Direct Mapping

Blocks in cache = 4

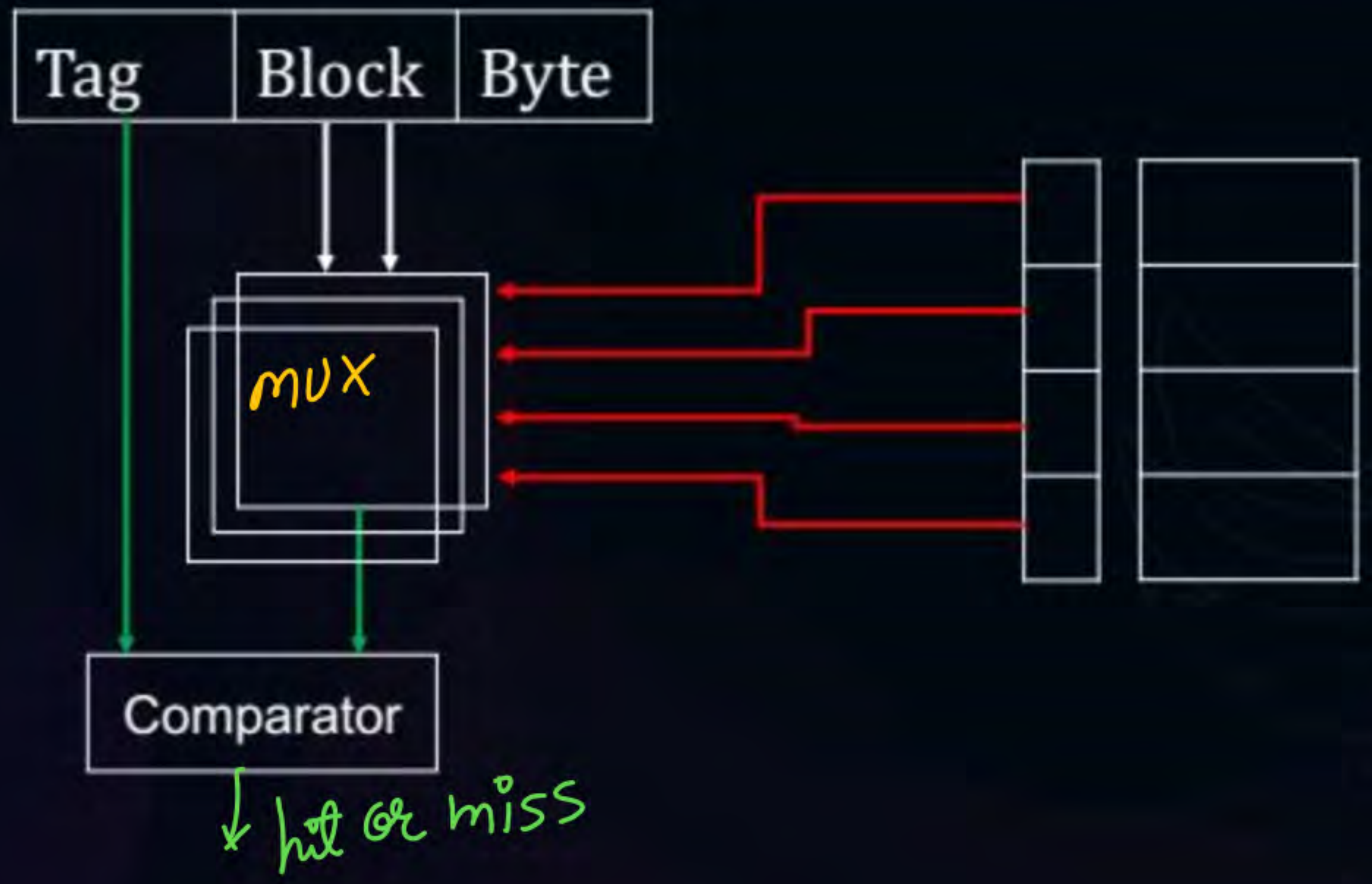
Block size = 16 bytes

Main memory address = 8-bits

Topic : Hardware Implementation in Direct Mapping



Topic : Hardware Implementation in Direct Mapping



Topic : Hardware Implementation in Direct Mapping

No. of MUX needed for tag selection = (No. of bits in tag)

Size of MUX —— " ———— = No. of blocks in cache : 1

No. of Comparators = 1

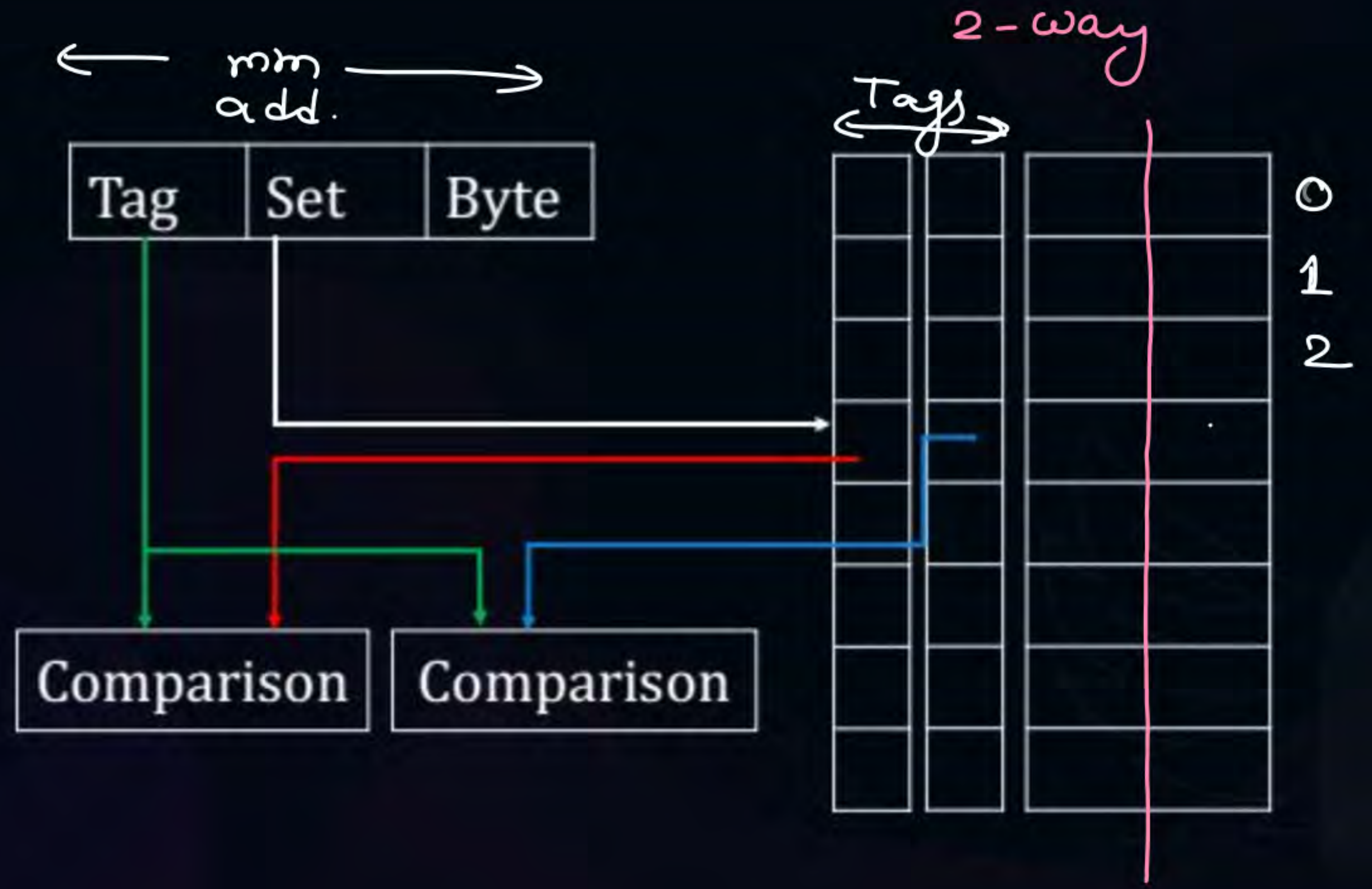
Comparator size = (no. of bits in tag) bits comparator

Topic : Hit Latency in Direct Mapping

Time needed by cache to check hit or miss

= mux delay for tag selection + Comparator delay

Topic : Checking Hit/Miss in Set Associative Mapping



Topic : Hardware Implementation in Set Associative Mapping

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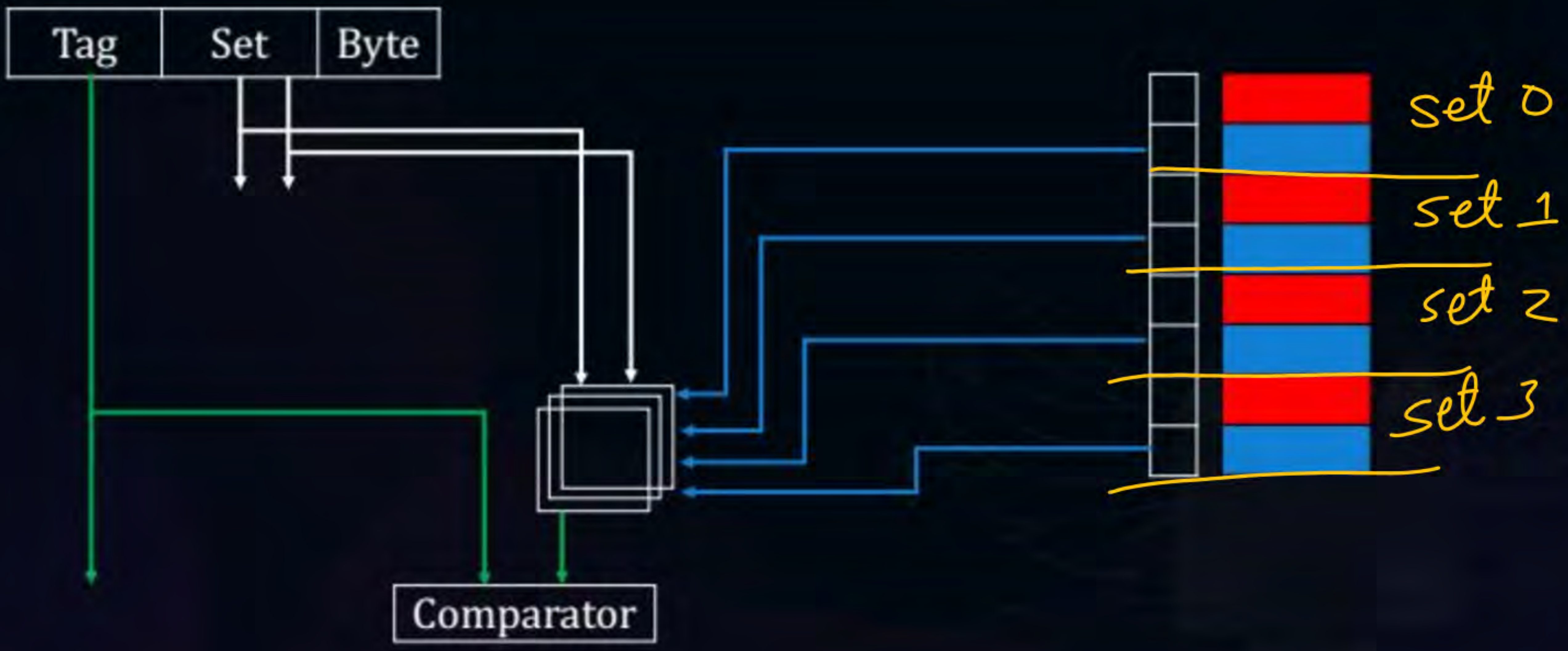
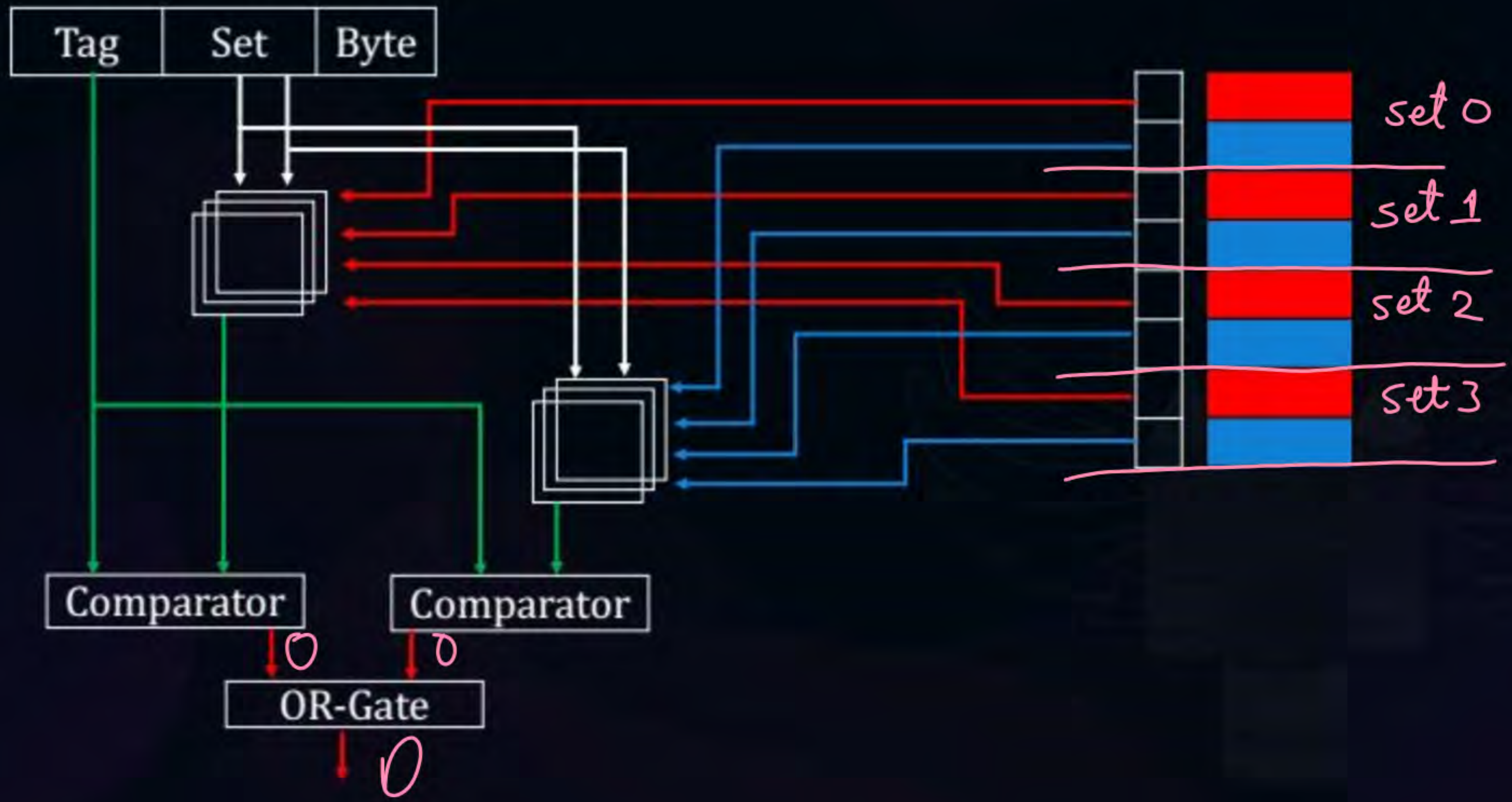


Topic : Hardware Implementation in Set Associative Mapping



Topic : Hardware Implementation in Set Associative Mapping

for k-way set associative cache

no. of MUX needed for tag selection = k * (no. of bits in tag)

size of —— || —————— = no. of sets in cache : 1

no. of comparators = k

size of comparators = (no. of bits in tag) bits comparator

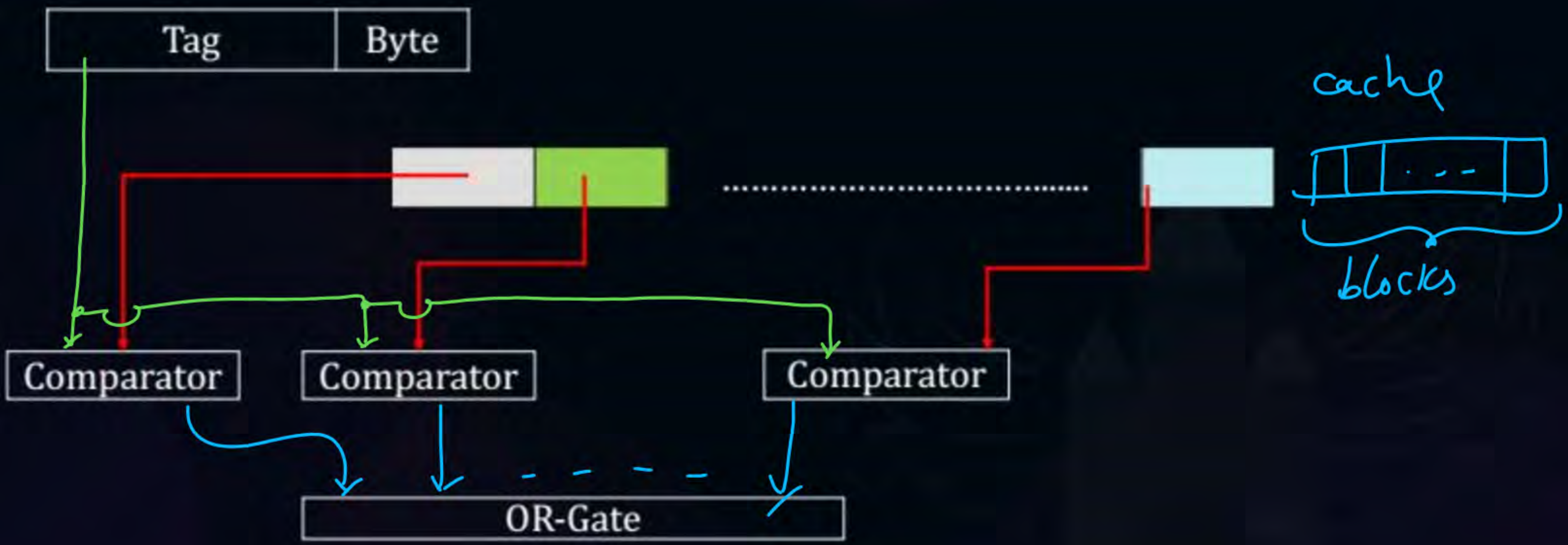
No. of OR gates = 1

no. of inputs in OR gate = k

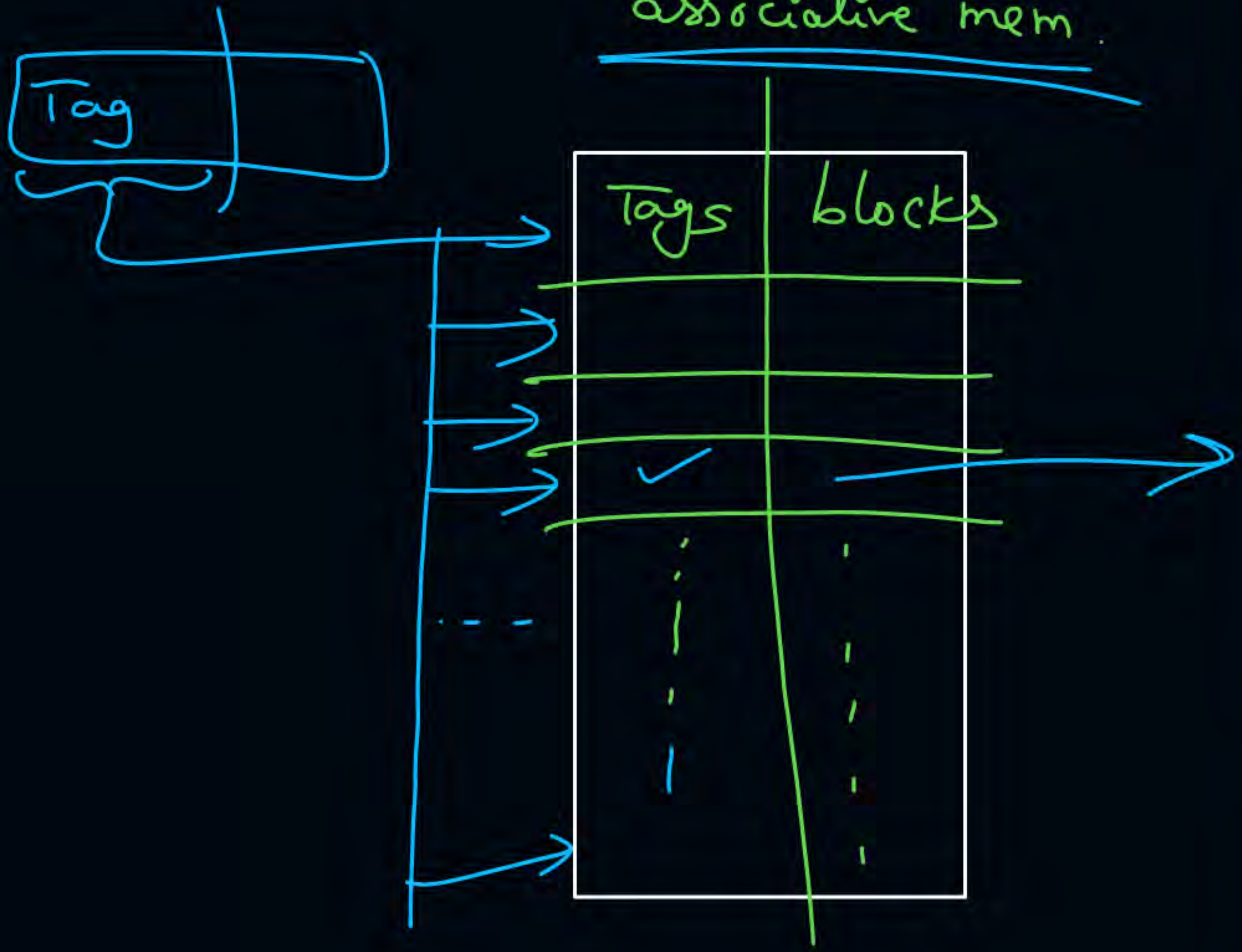
Topic : Hit Latency in Set Associative Mapping

= MUX delay for tag selection + comparator delay + OR gate delay

Topic : Checking Hit/Miss in Fully Associative Mapping



associative mem



Hardware :-

no. of comparators = no. of blocks in cache

size of comparator = (no. of bits in tag) bit comparator

1 OR gate of no. of inputs = no. of blocks in cache

Topic : Hit Latency in Fully Associative Mapping

= Comparator delay + OR gate delay

[NAT]

H.W.

#Q. Cache Size = 128KB

Block size = 32 bytes

Main memory address = 29-bits

4-way set associative cache

1. Tag size?
2. Tag Directory size?
3. Comparator required?
4. MUX required?

[MCQ]

H.W.

#Q. Consider two cache organizations. First one is 32 KB 2-way set associative with 32-bytes block size, the second is of same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k-bit comparator has latency of $\frac{k}{10}$ns. The hit latency of the set associative organization is h_1 while that of direct mapped is h_2.

The value of h_1 is:

A. 2.4 ns

B. 2.3 ns

C. 1.8 ns

D. 1.7 ns

[MCQ]

H.W.

#Q. Consider two cache organizations. First one is 32 KB 2-way set associative with 32-bytes block size, the second is of same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k-bit comparator has latency of $\frac{k}{10}$ ns. The hit latency of the set associative organization is h_1 while that of direct mapped is h_2.

The value of h_2 is:

A. 2.4 ns

B. 2.3 ns

C. 1.8 ns

D. 1.7 ns

2 mins Summary

Topic Miss Penalty

Topic Types of Misses

Topic Cache Mapping Hardware

Happy Learning

THANK - YOU